

DAY 5 – CASE STUDY: COMPONENT & HELPDESK TROUBLESHOOTING

Course: Diploma in Cyber Security

Topic: Multi-Hardware Incident
Diagnostics & Resolution

Name: HET ACHARYA

TASK 1 – LAPTOP NO-POWER INCIDENT DIAGNOSTIC PROCEDURE

Scenario: A laptop fails to power on even when connected to its AC power adapter

Step No.	Diagnostic Action	What I Am Checking For
1	Inspect AC Adapter & Wall Socket	Checking whether the AC wall outlet is supplying active current and verifying if the charger's LED indicator lights up.
2	Check Laptop Charging LED Indicator	Checking if the laptop's DC-in port detects incoming voltage and initiates battery charging.
3	Perform Hard Reset (Drain Residual Charge)	Disconnecting power, removing the battery (if external), and holding the power button for 30 seconds to drain lingering static charge from motherboard capacitors.
4	Cross-Test with a Known-Good Charger	Testing with a compatible original adapter to isolate a blown adapter fuse or damaged DC barrel pin.
5	Isolate Dead Battery vs Motherboard Failure	Attempting to boot the laptop directly on AC power without the battery connected to identify if a shorted battery is locking the power circuit.

ASK 2 – HELPDESK TICKET: BLANK SCREEN WITH RUNNING CPU

Helpdesk Ticket: *"Monitor displays a completely black screen, but PC fans and LEDs are active."*

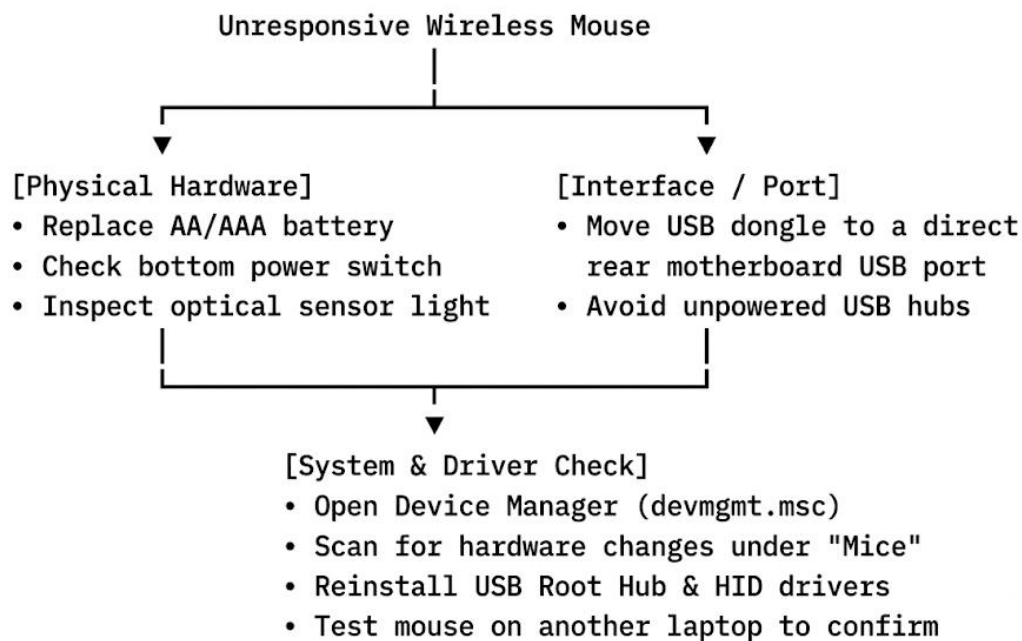
Diagnostic Workflow:

1. **Verify Monitor Power Status:** Ensure the monitor is switched on, the power cord is secure, and check if the power LED is blue/amber (standby).
2. **Check Display Signal Cable:** Unplug and firmly reseal the HDMI or DisplayPort cable on both the monitor and PC ends, inspecting for bent pins.

3. Verify Port Destination (Dedicated GPU vs Integrated):
Ensure the display cable is plugged directly into the dedicated graphics card port at the bottom, not the motherboard video output.
4. Input Source Selection: Use the monitor's physical OSD buttons to confirm the active input channel matches the connected port (e.g., HDMI 1 vs DisplayPort).
5. Display Output Reset Shortcut: Press Windows Key + Ctrl + Shift + B to restart the graphics driver subsystem, or press Windows Key + P to toggle display projection modes.
6. Internal Component Reseating: If the display remains blank, power down the PC, ground myself, and reseat the RAM modules and dedicated GPU in their PCIe slots.
7. Secondary Screen Isolation: Connect the desktop to an alternate monitor or TV to rule out a failed backlight or damaged panel.

TASK 3 – WIRELESS MOUSE ISOLATION MATRIX

Issue: Keyboard responds normally, but wireless mouse input is entirely unresponsive.



Mouse Fault: If replacing the battery and testing the mouse on another PC fails to trigger the optical LED or cursor movement, the internal optical sensor/transmitter is defective.

USB Port Fault: If the mouse immediately starts working when moved to a different USB port, the original port is loose or damaged.

System Setting Fault: If the mouse works on another machine but shows a yellow exclamation mark in Device Manager on this PC, the issue is an OS driver conflict.

TASK 4 – GAMING CAFÉ

OVERHEATING & THERMAL SHUTDOWN SCENARIO

Scenario: A high-end gaming PC continuously shuts down during intensive multiplayer gaming sessions due to thermal protection cut-off

Hardware Cause	Physical Check & Inspection Procedure	Corrective Action
1. Heavy Dust Accumulation on Heatsinks	Open the PC side panel and inspect the CPU cooler radiator, GPU fin stack, and case intake filters for thick dust blankets choking airflow.	Clear all heatsink fins and mesh filters using a can of dry compressed air and an anti-static brush.
2. CPU / GPU Fan Malfunction or Bearing Jam	Boot the system under load and visually verify that CPU, GPU, and case intake/exhaust fans spin freely at expected RPMs without rattling sounds.	Replace failed cooling fans or lubricate stiff bearings to restore case airflow.
3. Dried or Degraded Thermal Paste	Unmount the CPU cooler/AIO pump block to check if the thermal interface material has turned dry, chalky, or cracked.	Clean both mating surfaces thoroughly with 99% isopropyl alcohol and apply a fresh pea-sized dot of high-grade thermal compound.

CONCLUSION

In this practical case study, I addressed complex, multi-component hardware troubleshooting scenarios commonly encountered in enterprise and helpdesk environments. I established systematic isolation flows for power issues (differentiating between adapter, battery, and motherboard faults), display blackouts (verifying signal paths and display

drivers), and peripheral failures (isolating hardware, USB ports, and drivers). Finally, analyzing the gaming café thermal scenario reinforced how restricted airflow, failing fans, and degraded thermal paste trigger automatic thermal shutdowns to protect core processing hardware from permanent heat damage.